

Problem 3

Problem 3

Problem 1 Suppose $f(x) = \begin{cases} x^2 - ax & \text{if } x < 3 \\ a2^x + 7 + a & \text{if } x > 3 \end{cases}$

Find a so that $\lim_{x \rightarrow 3} f(x)$ exists.

Explanation. Remember, it's not enough to find a value. We must also justify that the limit exists for the value chosen.

To find a number a for which $\lim_{x \rightarrow 3} f(x)$ exists we find a such that $\lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^+} f(x)$.

To find the left-sided limit:

$$\lim_{x \rightarrow 3^-} f(x) = \lim_{x \rightarrow 3^-} (x^2 - ax) = 9 - 3a.$$

To find the right-sided limit:

$$\lim_{x \rightarrow 3^+} f(x) = \lim_{x \rightarrow 3^+} (a2^x + 7 + a) = a2^3 + 7 + a = 9a + 7.$$

In order for $\lim_{x \rightarrow 3} f(x)$ to exist we must have the two one-sided limits equal. That is,

$$\begin{aligned} \lim_{x \rightarrow 3^-} f(x) &= \lim_{x \rightarrow 3^+} f(x) \\ 9 - 3a &= 9a + 7 \\ 2 &= 12a \\ a &= 1/6. \end{aligned}$$